Can excitonic-insulator switches beat CMOS? A benchmarked equilibrium bound across the BEC–BCS crossover

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Exciton condensates and excitonic insulators have been proposed as switches that beat the $\ln(10)k_BT/e \approx 60$ mV/decade thermionic limit of complementary metal-oxide-semiconductor (CMOS) transistors, which would reduce the energy cost of computing. We test this premise for gate-driven devices whose ON and OFF states are in thermal equilibrium, using mean-field models with Gaussian fluctuations, each ingredient benchmarked against a published result: a two-dimensional (2D) electron–hole gap equation across the BEC–BCS crossover, superfluid-state screening validated against diffusion quantum Monte Carlo, Dirac bands for graphene double layers, a three-chain model of Ta_2NiSe_5 with exciton–phonon coupling, gate electrostatics and disorder. Screening plus the Nelson–Kosterlitz bound gives a ceiling $T_c \leq \mathrm{Ry}^*/4r_{s,c}^2$, where $r_{s,c}$ is the density at which the superfluid appears; no hBN-encapsulated double layer reaches 300 K. For graphene double layers at the bilayer pseudospin field-effect transistor operating point, the gap vanishes under both normal-state and superfluid-state screening. In Ta_2NiSe_5 the order is mostly excitonic (≈70–77%) but requires the lattice, and its discrete symmetry breaking precludes superfluidity. The only sub-CMOS switching found, an abrupt mean-field gap onset, does not survive fluctuations: the swing becomes 100–470 mV/decade at 300–350 K with an ideal gate, and at least 285 mV/decade at 300 K with gate electrostatics. Switching energy is 2–45 times that of ideal CMOS. A machine-learning surrogate finds no buildable design above ≈120 K with hBN dielectrics. The bottleneck for equilibrium excitonic logic is 2D phase stiffness under screening, not pairing. Tunnelling and non-equilibrium mechanisms are not assessed.

I. INTRODUCTION

Electricity use by data centres is projected to roughly double between 2024 and 2030, with artificial intelligence as the main driver[1]. A fundamental limit on the energy per logic operation in CMOS is the subthreshold swing $S \geq \ln(10)k_BT/e$, which sets the minimum gate voltage per decade of current, 60 mV/decade at room temperature[2]. Devices that sidestep this "Boltzmann tyranny" are actively sought.

Excitonic systems have been proposed in this context. An excitonic insulator (EI) forms when the single-particle gap E_g is smaller than the exciton binding energy E_b, and the ground state is a BCS-like condensate of electron–hole pairs[3,4]. In bilayers with spatially separated electrons and holes[5,6] the condensate supports counterflow superfluidity[5,7]. On this basis the bilayer pseudospin field-effect transistor (BiSFET) was proposed, with a simulated switching energy of 0.008 aJ per clock cycle, compared with ∼5 aJ for end-of-roadmap CMOS[8]. Its key assumption is a room-temperature condensate in graphene double layers, based on an unscreened mean-field estimate $T \sim 0.1E_F$[7]. Including carrier screening lowers this estimate by many orders of magnitude[9]. Experimentally, signatures of excitonic order exist in Ta_2NiSe_5[10–12], 1T-$TiSe_2$[13], monolayer WTe_2[14,15], TMD double layers[16,17], InAs/GaSb[18] and graphene double bilayers in the quantum Hall regime[19,20]. Room-temperature excitonic transistors exist, but they operate without a condensate[21].

This work asks a narrow question. Given the pairing physics as understood in the literature, can an excitonic switch beat CMOS at chip temperatures (300–350 K)? The approach is a chain of models in which each ingredient reproduces a published number before it is used (Table I), so the conclusion does not rest on unvalidated approximations. All code is openly available[22].

II. MODELS AND METHODS

A. 2D continuum gap equation

Electrons and holes with reduced mass m_r pair through $V(q) = 2\pi e^2 e^{-qd}/[\epsilon q(1 + r_0 q)]$, with layer separation d, background dielectric constant ϵ and Rytova–Keldysh length r_0[23]. The gap equation is

$$\Delta_k = \sum_{k'} V_{kk'} \frac{\Delta_{k'}}{2E_{k'}} \tanh \frac{E_{k'}}{2k_BT}, \qquad E_k = \sqrt{\xi_k^2 + \Delta_k^2}, \qquad \xi_k = \frac{E_g}{2} + \frac{\hbar^2 k^2}{4m_r}, \tag{1}$$

where $\mu_{\mathrm{eff}} = -E_g/2$ plays the role of the chemical potential in the BEC–BCS crossover[24]. The radial Nyström discretisation uses an analytic angular integral (complete elliptic integrals) and cell-averaging of the integrable logarithmic singularity. In 2D the mean-field T^* is only a pairing scale on the BEC side[24]. The condensation tem-

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perature is bounded by the Nelson–Kosterlitz jump[25], $k_B T_{\rm BKT} \le \pi\hbar^2 n_s/2M$ with $M = m_e + m_h$ and $n_s \le n$.

B. Screening in the superfluid state

At fixed pair density n [effective units a_B^* and Ry*; $r_s = (\pi n)^{-1/2}/a_B^*$; one species per layer[6]], the pairing interaction is screened by the carriers of the paired state [superfluid-state (SS) screening][26]:

$$V_q^{\rm SS} = \frac{v_q e^{-qd}}{1 + 2v_q\Pi(q) + [v_q\Pi(q)]^2(1 - e^{-2qd})}, \qquad (2)$$

with $v_q = 2\pi e^2/\epsilon q$, and $\Pi = \Pi_n + \Pi_a$ is built from BCS coherence factors, $\Pi(q) = \sum_k (u_k v_{k'} - v_k u_{k'})^2/(E_k + E_{k'})$ with $k' = k+q$. It reduces to the static 2D Lindhard function for $\Delta \to 0$ and vanishes as $q \to 0$ in the superfluid. Iteration starts from the unscreened (US) coherent state. Neilson *et al.* showed that this SS approach, unlike US or normal-state (NS) screening, reproduces the DQMC condensate fraction[26].

C. Dirac bands

For graphene double layers we use the two-band Dirac model[7], $\xi_k = \hbar v k - E_F$, coupling $\alpha = e^2/\epsilon\hbar v$, chirality factor $(1 + \cos\theta)/2 = [(k + k')^2 - q^2]/4kk'$ and four spin–valley copies. Three screening treatments are compared: US[7]; NS random-phase approximation[9]; and SS screening as in Eq. (2), with the intraband polarisability from coherence factors and the interband part kept at its normal-state value.

D. Quasi-1D chains: Ta_2NiSe_5

We use the three-chain Hubbard model of Kaneko *et al.*[4]. It has two Ta $5d$ conduction chains and one Ni–Se valence chain, with $t_c = -0.8$, $t_f = 0.4$ and $D = 0.2$ eV fitted to density-functional theory. It includes on-site U, interchain $V = U/4$, and a uniform shear phonon with $\lambda = g^2/\omega$. Since the mean-field equations of Ref. 4 are not given in its main text, we derive them from its Hamiltonian: Hartree terms for U and V, Fock terms on both c–f bonds, a static $q = 0$ phonon, and opposite order on the two Ta chains.

E. Fluctuations, gate and disorder

Gaussian fluctuations are included through a Ginzburg–Landau expansion of the Gibbs–Bogoliubov variational free energy along the mean-field pair shape, with stiffness $\hbar^2/2M$. The quartic term is treated in the self-consistent Hartree (large-N) approximation[27,28] with Bose occupation of the pair modes, $\langle|\psi|^2\rangle = \sum_Q n_B(a + b\langle|\psi|^2\rangle + cQ^2)$. Gaussian fluctuations about the normal state are equivalent to the Nozières–Schmitt-Rink scheme[24,29]. The fluctuation-broadened gap follows the pseudogap form $\Delta^2 = \Delta_{\rm sc}^2 + \Delta_{\rm pg}^2$ [30]. Here $\Delta_{\rm pg}$ is the pairing amplitude of non-condensed pairs. It measures local pairing, not long-range phase coherence, which in 2D at $T > 0$ is at most quasi-long-ranged[25]. The spectral gap entering the swing therefore requires pairing but not coherence. The fluctuation sum is evaluated strictly in 2D: it runs over in-plane pair momenta only, with no interlayer dispersion and no dimensional crossover. The NSR scheme was originally formulated in three dimensions[29], and only its equivalence to Gaussian fluctuations[24] is used here. In 2D, Gaussian fluctuations alone do not yield a finite-temperature condensate, so the self-consistent Hartree step keeps the pair density finite and the condensation temperature is taken from the BKT bound. The fluctuation analysis is applied only to the 2D double layers, not to the bulk quasi-one-dimensional Ta_2NiSe_5 (Sec. III D). Gate electrostatics follow Ref. 31: $e\,dV = (C + d\mu_x/dn)\,dn$ with $C = 4\pi e^2 d_s/\epsilon$. A thermal compressibility term $k_B T/n$ is added because the pair gas is non-degenerate at 300 K. Disorder enters through the Abrikosov–Gor'kov equation as applied to bilayers[32].

F. Device metrics

For a gap-switching transistor we assume intrinsic activated conduction, $G \propto e^{-E_{\rm gap}/2k_B T}$ [33]. The swing is then $S = 2\ln(10)k_B T/|dE_{\rm gap}/d(eV)|$, which is compared with $S_{\rm CMOS} = \ln(10)k_B T/e$. This treats the measured current as a direct probe of the thermodynamic gap. It assumes contacts in equilibrium with the channel and no parallel conduction paths. Sub-gap states, defect-mediated hopping and contact resistance would all make the measured S worse than this bound. Band-to-band tunnelling is a distinct mechanism that is not evaluated here. The mean-field overestimate of T_c by $(4e)^{1/3} \approx 2.2$ was derived for weak-coupling contact interactions in three dimensions[24,34]. Here it is used only as a heuristic robustness check on whether pairing survives at the operating temperature. The conclusions of Sec. III E hold without it, because the fluctuation analysis removes the sub-CMOS swing independently. For the BiSFET, the criterion is a 2D condensate at the operating temperature. Switching energy at equal geometry scales as $(S_{\rm EI}/S_{\rm CMOS})^2$. As a sanity check, $C_g W L V^2$ for the BiSFET geometry gives 0.0044 aJ, against the SPICE value of 0.008 aJ[8].

TABLE I. Benchmarks reproduced before each ingredient is used; all rows are regenerated by `run_benchmarks.py`[22]. Literature values read from figures are approximate ($\approx \pm 0.03$). Two Ta_2NiSe_5 rows are documented deviations (see text).

ingredient	quantity	literature	this work
2D gap equation	2D exciton $E_b = 4\,\mathrm{Ry}^*$	435.4 meV (exact)	432.9 meV
	EI exists only for $E_g < E_b$	Ref. 3	yes
crossover mean field	3D unitarity Δ/E_F, μ/E_F	0.68, 0.59[24]	0.686, 0.591
	BCS-limit T_c	$(8\gamma/\pi e^2)E_F e^{-\pi/2k_F\lvert a\rvert}$[24]	within 1%
fixed-density bilayer	$\Delta_{\max}$ at $d = a_B$, $r_s = 1.5, 3, 20$	0.66, 0.68, 0.18 Ry*[6]	0.659, 0.700, 0.173
SS screening	superfluid onset $r_{s,c}$, $d = a_B$	2.5 (DQMC), ≈ 3 (SS)[26]	2.38
	onset, $d = 0.7a_B$	2.1 (DQMC)[26]	1.97
Dirac, NS screening	$V_{\mathrm{us}}(p_F)/V(p_F)$, $\alpha = 1$	≈ 9[9]	8.99
	$\lambda_c^{\max}$	$1/16$[9]	0.0624
	$d\lambda/d\ln(1/T)$ vs. λ_c ($\alpha = 1$)	equal[9]	0.0546 vs. 0.0567
Ta_2NiSe_5 chains	distortion onset λ_c	0.125 eV[4]	0.125 eV
	EI onset V_c	0.6 eV[4]	0.69 eV (+15%)
	k_BT_c ($V = 0.55$, $\lambda = 0.04$ eV)	0.0783 eV[4]	0.0564 eV (−28%)
gate	interlayer E_b, $d = a_B$	0.77 Ry* (linear fit)[31]	0.866 Ry* (exact)
disorder	$dT_c/d\delta$; critical δ	$-\pi/4$; $0.88T_{c0}$[32]	-0.7855; $0.8819T_{c0}$
energy	BiSFET switching energy	0.008 aJ[8]	0.0044 aJ (CV^2)

III. RESULTS

A. Crossover and the pairing–condensation gap

Figure 1 shows the 2D crossover for $MoSe_2$/hBN/WSe_2-like parameters ($m_r = 0.22$, $\epsilon = 4.5$, $d = 1.6$ nm). On the BCS side, Δ_k peaks at the Fermi wavevector and v_k^2 has a Fermi edge. On the BEC side, both are smooth. With the unscreened interaction, the mean-field pairing scale T^* exceeds 350 K over most of the EI window, but the BKT bound keeps the condensation temperature below ≈200 K, and to only tens of kelvin near the crossover. Screening (Sec. III B) removes the high-density end. This separation of T^* and T_c on the BEC side mirrors the cold-atom phase diagram[24]. It is the first sign that the bottleneck is phase stiffness, not pairing.

B. Screening ceiling

With superfluid-state screening, a large-gap superfluid exists only for $r_s > r_{s,c}(d/a_B)$, in agreement with diffusion quantum Monte Carlo [Fig. 2(a)]. Below $r_{s,c}$, screening suppresses the gap exponentially[26]. Combining this with $n_s \le n$ gives

$$k_BT_c \le \frac{\mathrm{Ry}^*}{4r_{s,c}^2}\,\frac{4m_r}{m_e + m_h}, \tag{3}$$

where the equal-mass case is the most favourable. With the $MoSe_2$/WSe_2 parameters of Ref. 31 ($m_r \approx 0.2$, $\epsilon \approx 5$, $d = a_B \approx 1.3$ nm) the bound is 57 K. With the representative values used in Figs. 1 and 4 ($m_r = 0.22$, $\epsilon = 4.5$, $d = 1.6$ nm) it is 49 K, and at the thinnest barrier, $d = 0.7$ nm, it is 112–121 K. Across $0.05 \le m_r \le 0.6$, $2 \le \epsilon \le 15$ and $0.6 \le d \le 3$ nm, no point reaches 350 K, and none with $\epsilon \ge 4.5$ reaches 300 K [Fig. 2(b)]. The maximum is 328 K, reached at $\epsilon = 2$. This result is unchanged when $r_{s,c}$ is extrapolated linearly below the smallest computed separation, $d/a_B = 0.3$. Valley and spin degeneracy lower the bound further: more degenerate channels increase the polarisability and screen the pairing interaction more strongly, so the effective coupling falls. In the large-N limit it scales as $1/N$, where N is the number of fermion species[9].

C. Graphene double layers: the BiSFET operating point

At the BiSFET density ($n = 4.9 \times 10^{12}$ cm^{-2}, $E_F = 258$ meV), with $d = 1$ nm and $\epsilon = 3.9$, the unscreened Dirac gap is $\Delta = 31$ meV ($\approx 0.12E_F$), consistent with Ref. 7. This corresponds to $T_{\mathrm{KT}} \le E_F/8 \approx 374$ K. With normal-state screening the coupling is $\lambda_c < 1/16$ and $T_c \sim 10^{-7}E_F$[9]. With superfluid-state screening the gap collapses within two iterations. Even at $\epsilon = 1$ and $d = 0.3$ nm it converges to 0.087 meV. Monolayer-graphene double layers therefore do not condense at useful temperatures under either screening treatment validated in the literature. This agrees with the conclusion of Ref. 26 for monolayer graphene at 1 nm separation.

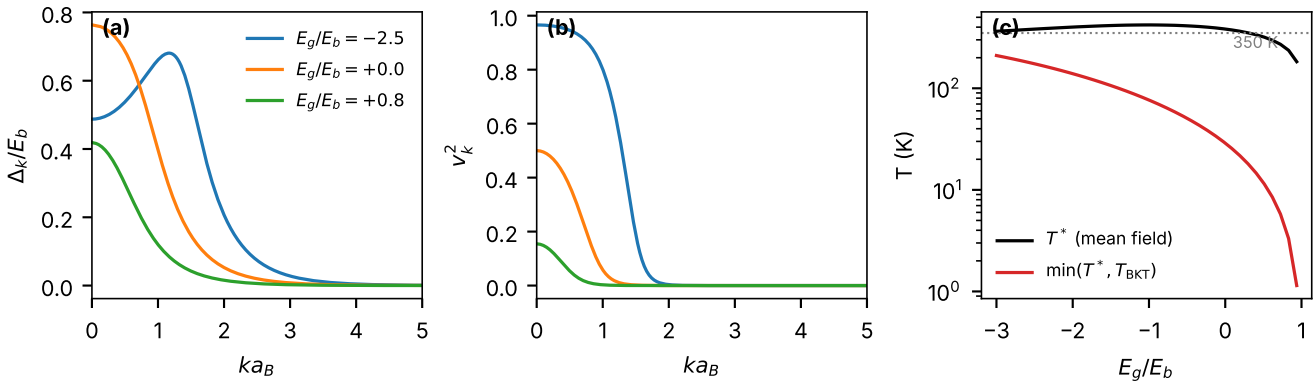


FIG. 1. BEC–BCS crossover in a 2D e–h double layer with the unscreened interaction ($m_r = 0.22$, $\epsilon = 4.5$, $d = 1.6$ nm; representative $MoSe_2$/hBN/WSe_2 values). (a) Gap function, (b) pair occupation, (c) mean-field pairing scale T^* and condensation estimate $\min(T^*, T_{BKT})$ versus E_g/E_b.

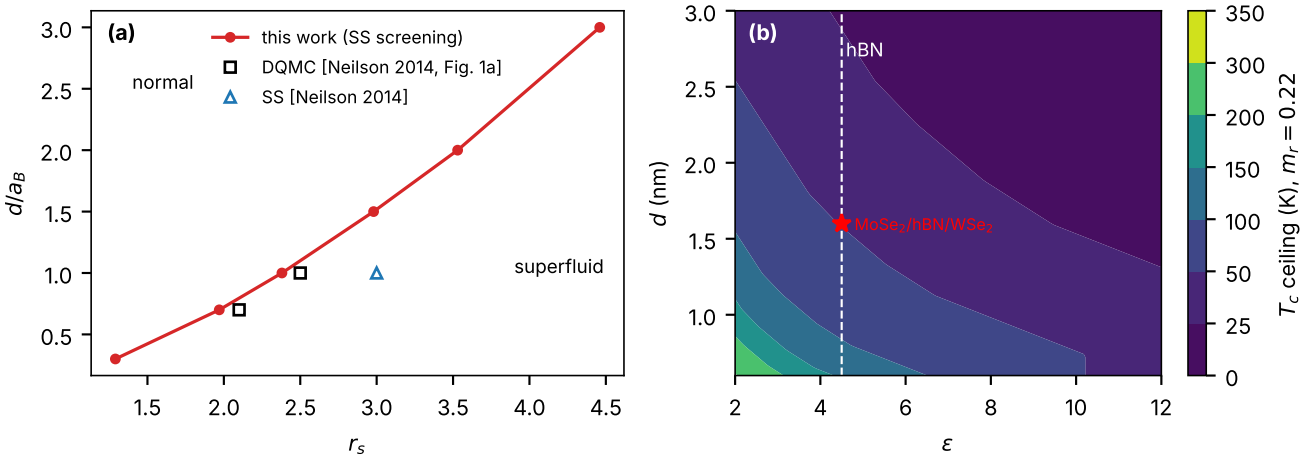


FIG. 2. (a) Superfluid onset $r_{s,c}$ versus layer separation with superfluid-state screening, compared with DQMC and mean-field SS points read from Ref. 26. (b) Screening ceiling, Eq. (3), for $m_r = 0.22$. The white contour marks 300 K and the dashed line marks hBN ($\epsilon = 4.5$).

D. Ta_2NiSe_5

In the three-chain model (Fig. 3) the order is cooperative. At $V = 0.55$ eV, below the purely excitonic threshold, order appears only for $0.02 < \lambda < 0.04$ eV or larger. For $\lambda = 0.04$–0.05 eV, 69–74% of the c–f hybridisation is excitonic, and 77% with the alternative decoupling discussed below. A purely structural instability would need $\lambda \geq 0.125$ eV, 2.5–6 times the estimate $\lambda \approx 0.02$–0.05 eV inferred from the measured distortion[4]. Our implementation reproduces the published distortion threshold exactly, but it places the purely excitonic threshold 15% higher ($V_c = 0.69$ vs. 0.6 eV) and $k_B T_c$ 28% lower than Ref. 4 (Table I). Two alternative Fock decouplings bracket both published values, so the deviation reflects mean-field decoupling details not specified in the main text of that work. The thresholds are therefore sensitive to the decoupling at the 15–30% level, whereas the excitonic share of the order is not: it stays at 74–77% across decouplings at $V = 0.55$ eV, $\lambda = 0.04$ eV. We therefore claim only that the order is predominantly excitonic, not a precise ratio. The device conclusion for Ta_2NiSe_5 below rests on the symmetry analysis of Ref. 35, not on these numbers. The mean-field T_c divided by 2.2 gives 298–377 K, which brackets the measured 326 K[11]. Symmetry analysis based on first-principles Wannier models shows that the transition breaks only discrete symmetries and that there is no excitonic superfluidity[35]. The symmetry argument is as follows. Symmetry-allowed single-particle hybridisation between the Ta and Ni states explicitly breaks the continuous U(1) symmetry associated with the conservation of relative electron–hole number[35]. The excitonic order then breaks only discrete lattice symmetries and is Ising-like. It has no gapless Goldstone mode,

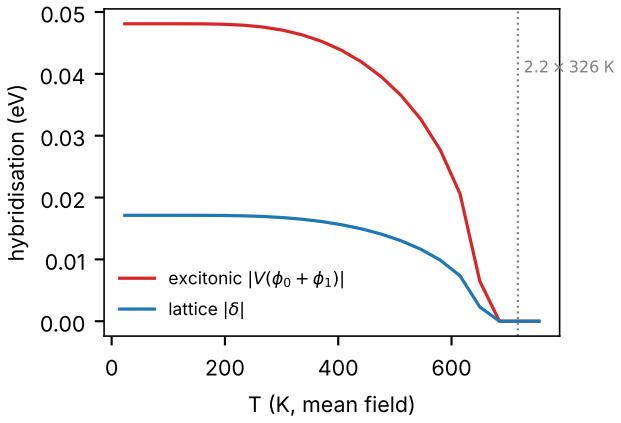


FIG. 3. Ta_2NiSe_5 three-chain model[4] at $V = 0.55$ eV, $\lambda = 0.04$ eV: excitonic and lattice contributions to the c–f hybridisation versus mean-field temperature.

its collective modes are gapped, and it cannot carry a supercurrent[35]. The same conclusion, that interband transitions fix the phase of the order parameter and make superfluidity impossible in a homogeneous excitonic insulator, was reached early on by Lozovik and Yudson[5]. The structural interpretation of Ref. 36 is compatible with our finding that the lattice decides whether order sets in. For devices, Ta_2NiSe_5 cannot host a counterflow condensate, and as a bulk crystal its gap can be tuned only by pressure or strain.

E. Switching steepness and energy

The only sub-CMOS swing found in mean field comes from an abrupt gap onset where $T^*(E_g)$ crosses the operating temperature: 60.4 mV/decade at 350 K for $MoSe_2$/hBN/WSe_2 and 38.6 mV/decade at 300 K for WTe_2 (Fig. 4). Physically, thermal fluctuations of the pairing amplitude persist above the mean-field T^* as a non-condensed pair gas, so the gap opens gradually rather than at a sharp transition. Gaussian fluctuations thus replace the onset with a smooth pseudogap crossover, giving $S = 102$–140 and 137–280 mV/decade respectively across a factor-of-four range of momentum cut-off; over both materials at both temperatures the range is 100–470 mV/decade. The Ginzburg–Landau quartic reproduces the mean-field amplitude to within 4%. A real gate makes the situation worse. At 300 K, interlayer charging plus the thermal compressibility of the pair gas give $S \geq 285$ mV/decade. A negative exchange compressibility[31] alone would give 49 mV/decade, but this gain does not survive at non-degenerate densities. At equal geometry, the switching energy is 2.2–4.1 times ($MoSe_2$/hBN/WSe_2) and 4.5–45 times (WTe_2) that of ideal CMOS at 350 K. Disorder is not limiting at room temperature: preserving $T_{c0} = 350$ K requires only a pair-breaking time $\tau > 12$ fs[32].

F. Machine-learning screen

A bootstrap ensemble of gradient-boosted trees[37] was trained on 3000 solver labels $T_c = \min(T^*/2.2, T_{\mathrm{BKT}}, \text{Eq. (3)})$, with $0.01 \leq m_r \leq 1$, mass ratio 0.2–5, $1.5 \leq \epsilon \leq 15$ and $0.3 \leq d \leq 10$ nm. On held-out data it reaches $R^2 = 0.87$ with a mean absolute error of 1.3 K, and 95% of held-out points lie within the ensemble's own $\pm 2\sigma$ predictive interval, so its uncertainty estimates are calibrated. Inverse design over 2×10^5 candidates, followed by verification with the full solver, gives at most ≈ 117 K for $\epsilon \geq 4.5$ and ≈ 233 K for $\epsilon \approx 1.6$. One label of 3000 exceeds 350 K, at $\epsilon = 1.7$, $d = 0.58$ nm and $E_g = -1.4$ eV, which is not a realistic device. The surrogate only interpolates the benchmarked model. It cannot discover mechanisms the model does not contain, so the inverse design bounds what this framework allows, not what all materials allow.

IV. ROBUSTNESS, SCOPE AND LIMITATIONS

Several ingredients that could raise T_c were treated optimistically: equal masses, $n_s = n$, an ideal gate in the bound, and single-species screening. The main conclusions were tested against the most consequential approximations as follows.

Screening model. Using the DQMC onset ($r_{s,c} \approx 2.5$ at $d = a_B$[26]) instead of the SS mean field gives 51 K for the $MoSe_2$/WSe_2 parameters of Ref. 31. DQMC includes dynamic screening and vertex corrections[26]. Reaching 300 K there would require $r_{s,c} \leq 1.03$, less than half of both the SS and DQMC values. In the most favourable hBN corner of the parameter box ($\epsilon = 4.5$, $d = 0.6$ nm), the ceiling is 146 K. It would reach 300 K only if the true onset lay ≈30% below the DQMC value. This is the margin by which the hBN conclusion depends on the screening physics.

Intralayer exchange. It enters the compressibility $d\mu_x/dn$ and hence the gate response, which we bracket using the exchange-dominated value of Ref. 31. At 300 K the thermal compressibility k_BT/n dominates both brackets. The ceiling of Eq. (3) does not involve the compressibility.

Ta_2NiSe_5. The thresholds depend on the mean-field decoupling; the excitonic share and the absence of superfluidity do not (Sec. III D).

Scope. Only equilibrium, gate-driven operation is evaluated. The main limitations are: (i) static mean field with Gaussian fluctuations, without vortex dynamics or vertex corrections beyond those tested in Ref. 26; (ii) no intralayer exchange in the continuum model, which changes the sign of the exciton–exciton interaction compared with Ref. 31; we use both as brackets; (iii) a 15–

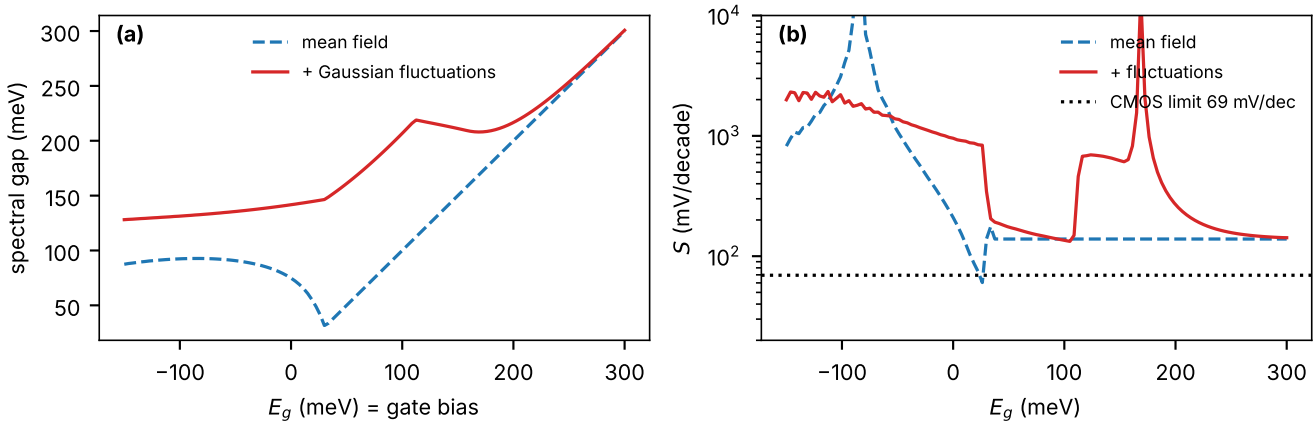


FIG. 4. Gap-switching transistor for $MoSe_2$/hBN/WSe_2 at 350 K. (a) Spectral gap versus E_g (ideal gate), in mean field and with Gaussian fluctuations. (b) Subthreshold swing, compared with the CMOS limit.

28% deviation from two Ta_2NiSe_5 numbers of Ref. 4. Its full mean-field equations are not given in its main text and were derived here independently, and two alternative Fock decouplings bracket the published values; and (iv) no transport, contact or circuit modelling. Non-equilibrium (optical) switching in Ta_2NiSe_5 and cryogenic operation, where the CMOS limit falls to 15 mV/decade at 77 K, lie outside this screen. Extending the bound to driven systems would require time-dependent treatments of the order parameter coupled to the pump field, such as time-dependent mean-field or non-equilibrium Green's-function approaches, together with transport calculations that include the contacts and the energy cost of the drive. The relevant comparison is between the switching or pump time scale and the thermalisation and recombination times of the electron–hole system. If thermalisation is fast compared with the switching cycle, the device passes through quasi-equilibrium states and the present bound applies at each instant. Only when the drive outpaces thermalisation can the constraints derived here be evaded.

V. CONCLUSION

Within benchmarked equilibrium models of 2D parabolic, Dirac and quasi-one-dimensional excitonic systems, gate-driven excitonic-insulator switches do not beat CMOS in subthreshold swing or switching energy at 300–350 K. Tunnelling-based and non-equilibrium (e.g. optically driven) excitonic switches are not ruled out by this analysis. Room-temperature pairing is possible, but screening-limited 2D condensation is not. The steep switching that appears in mean field is an artefact that Gaussian fluctuations remove. The design rule of Eq. (3) and the open code provide a quantitative test for future excitonic device proposals.

DATA AVAILABILITY

All code, data and figure scripts are openly available at `https://github.com/masyhur-n/excitonic-insulator-sim`[22] under the MIT license, and archived at Zenodo, doi:10.5281/zenodo.23684703[22].

AUTHOR DECLARATIONS

Conflict of interest. The author has no conflicts to disclose.
Author contributions. Nailah Masyhur: conceptualization, methodology, software, validation, formal analysis, writing (original draft, review and editing).
Ethics approval. Not applicable.
Use of AI tools. Code development, literature retrieval and drafting were assisted by an AI coding assistant (Claude, Anthropic). The author directed the study, verified every result against the published benchmarks listed in Table I, and takes full responsibility for the content.

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